Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/15/2025
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Novel Explorations Into The Interactions Between Light and Gravity Sensing in Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Organization Name:	University of North Carolina-Greensboro		
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City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/10/2017	**End Date:**	06/30/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 6/30/2024 per F. Hernandez/ARC (Ed., 7/5/23) NOTE: End date changed to 6/30/2023 per NSSC (Ed., 5/16/22) NOTE: End date changed to 6/30/2022 per F. Hernandez/ARC (Ed., 8/19/21) NOTE: End date changed to 7/31/2021 per F. Hernandez/ARC (Ed., 6/4/2020) NOTE: Extended to 7/31/2020 per NSSC information (Ed., 5/15/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schug, Malcolm (UNC Greensboro)		
Grant/Contract No.:	80NSSC17K0546		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	[ED. NOTE 12/11/2017: Continuation of "Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants--NNX12AO65G," per PI move from University of Mississippi to University of North Carolina at Greensboro.] The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the International Space Station (ISS) with the European Modular Cultivation System (EMCS) using the model plant Arabidopsis. In this new proposed project, we plan to confirm and extend our discovery during the recent ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Our hypothesis is that positive red-light-sensing, which was known in older plant lineages, is masked by normal 1-g conditions in more recently evolved lineages. We also propose to investigate fundamental interactions among red and blue light signaling pathways and the gravity sensing mechanisms. The ISS is the only laboratory facility where it is possible to reliably and simultaneously vary light and gravity stimuli for the proposed experiments. These experiments can be accomplished with telemetric science and therefore fit with the extremely limited down mass capabilities of current spaceflight opportunities. The EMCS is an automated facility and can be operated largely with telemetry. Experimental containers will be launched via Space-X, and during an experimental run, images of seedling growth responses can be downlinked to Earth. A significant advantage of this proposed project is that we will use existing flight hardware that has been flown successfully and allowed for growth of Arabidopsis seedlings in the EMCS. This project is also relevant to the recent emphasis at NASA to study plant growth and development at fractional g-levels such as those found on the moon and Mars. Improved knowledge of the basic mechanistic processes that will be the focus of this project is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying the effects of gravity and light on basic cellular mechanisms in plants, this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology. Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our spaceflight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture. In addition, improved knowledge of light and gravity systems in plants will, in the long term, allow us to develop better crop plants for use on Earth.
Task Progress:	Using the European Modular Cultivation System (EMCS), it is possible to study plant growth and development at fractional or reduced gravity levels as would be seen on the Moon or Mars. Improving the knowledge base of these basic processes is important as techniques are developed to use plants in bioregenerative life support systems for future manned missions. In the long term, a better understanding of how plants are affected by these factors helps in the development of strategies to optimize light sensing in plants and modify plant species by using biotechnological approaches to improve crop plants. To our knowledge, for the first time in the course of plant space biology, we have information on a physiological response (phototropism) across a continuum of gravity conditions (0.5g, 0.8g, µg, 0.1g, 0.3g, 1.0g). The final outcome of any scientific project is a peer-reviewed publication. We are pleased to report that 22 papers resulted from NASA funding this project. [Ed. Note: See Cumulative Bibliography for this investigation; grant numbers 80NSSC17K0546, NNX12AO65G, and NNX10AM86G.]
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Articles in Peer-reviewed Journals	Kiss JZ. "Statoliths function in gravity perception in plants: Yes, no, yes!" Planta. 2025 Jan 27;261(2):45. Review. https://doi.org/10.1007/s00425-025-04631-2 ; PMID: 39869172 , Jan-2025
Articles in Peer-reviewed Journals	Hughes AM, Jenkins BE, Bauer LV, Kiss JZ. "Performance and accuracy of the automated measurement software: Simple Online Automated Plant Phenomics (SOAPP)." Gravit Space Res. 2025 Aug 7;13(1):51-64. https://doi.org/10.2478/gsr-2025-0004 , Aug-2025